

GROUP IT, MAP IT! SETS & VENN DIAGRAMS

Grade 5 Mathematics Curriculum Companion & Interactive Worksheet

Topic: Set Theory & Venn Diagrams

Grade: 5th Grade

Subject: Mathematics

1 Ready for the Picnic!

Our Story Begins...

Ananya and Kabir are packing for their school picnic to Kanha National Park. They have lots of snacks, games, and supplies, but everything is completely jumbled up! Can you help them sort and group their things? Let's get packing!

Sorting Indian Street Food!

Help Ananya and Kabir sort their favorite Indian street foods:

List of Snacks: *Samosa, Murukku, Pani Puri, Banana Chips, Pakora, Mirchi Bajji, Kolhapuri Misal.*

Think & Sort: Which snacks are CRISPY? Which are SPICY? Can some snacks be BOTH?

2 What is a Set?

In mathematics, a **Set** is a collection of well-defined objects that belong together based on a common rule or property.

Key Rule: You know exactly what belongs in a set and what does not!

Set Name	Elements in the Set	Common Property / Rule
Rohan's Geometry Box	{Triangle, Circle, Square, Rectangle, Ruler}	Tools used for geometry.
Traditional Indian Breakfast	{Idli, Dosa, Poha, Paratha, Upma}	Popular Indian breakfast items.
South Indian States	{Tamil Nadu, Kerala, Karnataka, Andhra Pradesh, Telangana}	The 5 states located in South India.

3 Overlapping Sets & Venn Diagrams

A **Venn Diagram** uses overlapping circles to visually organize items and show relationships between different sets.

Case Study: Crispy vs. Spicy Snacks

- **Crispy Snacks Set:** {Samosa, Murukku, Banana Chips, Pani Puri, Pakora, Kolhapuri Misal}
- **Spicy Snacks Set:** {Mirchi Bajji, Kolhapuri Misal, Pani Puri, Pakora}
- **Both Crispy & Spicy (Overlap):** {Pani Puri, Pakora, Kolhapuri Misal}

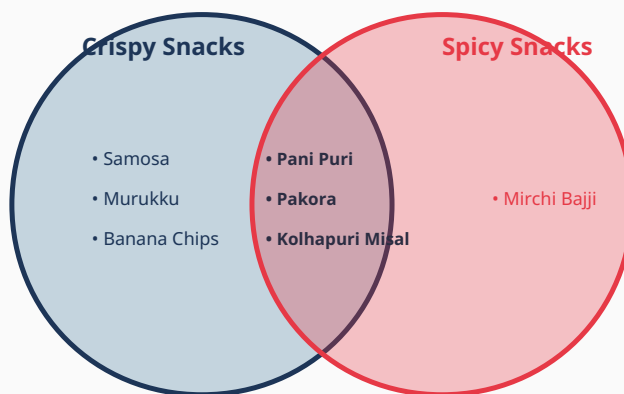


Figure 1: Venn Diagram showing Crispy, Spicy, and Overlapping (Both) Snacks

4 Key Operations: Union & Intersection

Concept	Definition	Example from Snacks
Union of Sets ($\cup$)	ALL items in Set A OR Set B (or both). Combines everything into one single group.	{Samosa, Murukku, Banana Chips, Pani Puri, Pakora, Kolhapuri Misal, Mirchi Bajji}
Intersection of Sets ($\cap$)	Items that belong to BOTH Set A AND Set B at the same time (the overlapping region).	{Pani Puri, Pakora, Kolhapuri Misal}

5 Advanced Set Concepts

Disjoint Sets (No Overlap)

Some sets have NO common elements—they don't overlap at all! These are called **Disjoint Sets**.

- **Railway Vehicles:** {Train, Metro, Tram, Monorail}
- **Flying Vehicles:** {Aeroplane, Helicopter, Hot Air Balloon}



Figure 2: Disjoint Sets with zero overlap

The Universal Set

The Universal Set (U) includes ALL objects being considered in a specific problem or context.

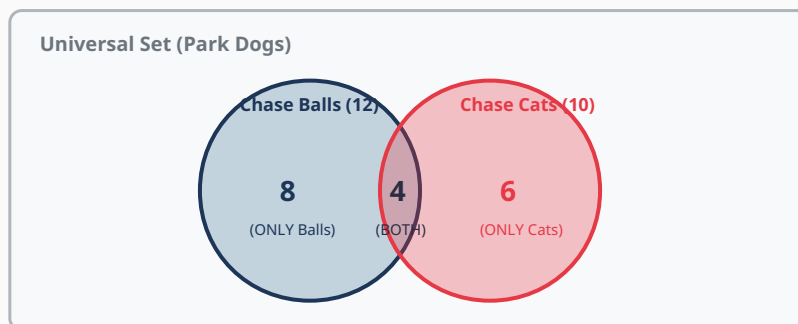
Example: All 30 students in Class 5A love ice cream!

- **Universal Set (U):** All 30 students in Class 5A.
- **Subsets:** Mango Kulfi lovers and Chocolate Ice Cream lovers.
- **Overlap:** Students who love BOTH Mango Kulfi and Chocolate!

6 Problem-Solving with Venn Diagrams

Example Problem: Dogs Chasing Balls & Cats

Problem Statement: In a park, 12 dogs chase balls, 10 dogs chase cats, and 4 dogs chase both. How many dogs chase ONLY balls?



Step-by-Step Solution:

- Total dogs chasing balls = 12
- Dogs chasing BOTH = 4
- Dogs chasing ONLY balls = Total chasing balls – Dogs chasing both = $12 - 4 = 8$ dogs

Answer: Exactly 8 dogs chase only balls!

GROUP IT, MAP IT! PRACTICE WORKSHEET

Grade 5 - Sets & Venn Diagrams

Student Name: _____

Date: _____

Score: _____ / 20

Question 1: Definition of a Set (3 Marks)

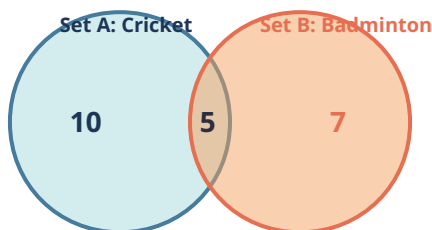
Which of the following represents a **well-defined set**?

- ☐ A) A collection of the most delicious snacks in India.
- ☐ B) The collection of states in South India ({Tamil Nadu, Kerala, Karnataka, Andhra Pradesh, Telangana}).
- ☐ C) A group of very tall students in Class 5.
- ☐ D) A collection of fun games to play at a picnic.

Explanation: Write 1-2 sentences on why your answer is well-defined:

Question 2: Reading a Venn Diagram (4 Marks)

Look at the Venn diagram below showing students' sports preferences in Class 5 (**Set A: Cricket**, **Set B: Badminton**).



a) What does the middle overlapping region with the number 5 represent?

b) How many total children play Cricket? (Hint: Remember to include the overlap!)

Question 3: Disjoint Sets Identification (4 Marks)

Explain why the set of **Railway Vehicles** ({Train, Metro, Tram}) and the set of **Flying Vehicles** ({Aeroplane, Helicopter, Hot Air Balloon}) are called **Disjoint Sets**.

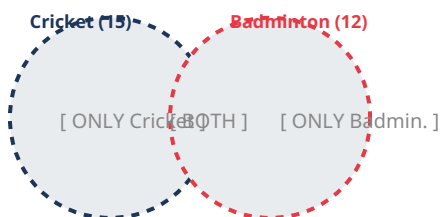
Draw a quick sketch of how their two circles should look in the box below:

[Draw Disjoint Venn Diagram Here]

Question 4: Word Problem & Diagram Mapping (5 Marks)

Sports in the Colony: In a neighborhood colony, 15 children play Cricket, 12 children play Badminton, and 5 children play BOTH Cricket and Badminton.

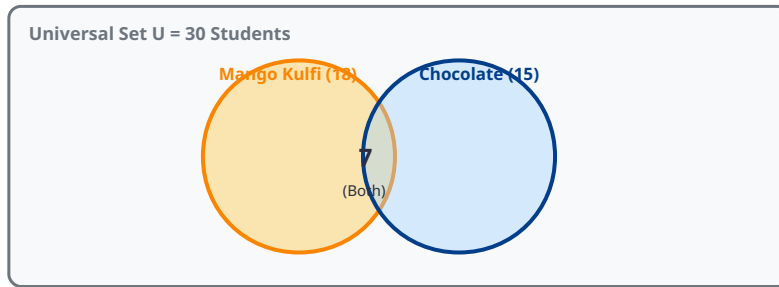
Fill in the Venn diagram template below with the correct numbers, then answer the questions:



- a) How many children play **ONLY Cricket**? _____
- b) How many children play **ONLY Badminton**? _____
- c) What is the total number of children who play at least one of these sports (the **Union** of both sets)?

Question 5: Universal Set & Ice Cream Survey (4 Marks)

In a class of 30 students (**Universal Set U**), 18 students like Mango Kulfi, 15 like Chocolate Ice Cream, and 7 like BOTH.



- a) How many students like **ONLY Mango Kulfi**? _____
- b) How many students like **ONLY Chocolate Ice Cream**? _____
- c) How many students like **NEITHER** Mango Kulfi nor Chocolate Ice Cream? (Hint: Subtract the total students who like at least one flavor from the Universal Set of 30)
- _____